

CLOUD ENGINEERING BOOTCAMP · DAY 3

Networking Fundamentals

for Cloud Engineers

From how google.com loads to how AWS VPCs are built — the networking model every cloud engineer relies on.

DURATION

3 hours

PREREQUISITES

Day 1: Cloud Basics · Day 2: Linux & SSH





Learning Objectives

By the end of today, you will be able to:



Explain how computer networks work



Understand the OSI & TCP/IP models



Explain IPv4 addressing & subnetting



Identify Public vs Private IP addresses



Understand CIDR notation



Explain ports and protocols



Understand DNS and DHCP



Explain routers, switches, NAT & firewalls



Understand VPNs and encrypted tunnels



Troubleshoot basic network issues



Map today's concepts directly to AWS VPC



Today's Agenda

1

Networks, Devices & Models (OSI / TCP-IP)

2

IP Addressing, Subnetting & CIDR

3

Routing, Switching, Ports & Protocols

4

DNS, DHCP, NAT, Firewalls & VPNs

5

Troubleshooting & Hands-On Lab

6

AWS VPC Mapping & Wrap-Up

Let's Warm Up



“What actually happens when your computer opens google.com?”

No wrong answers — we'll build the real, complete picture together today.



What Is a Computer Network?

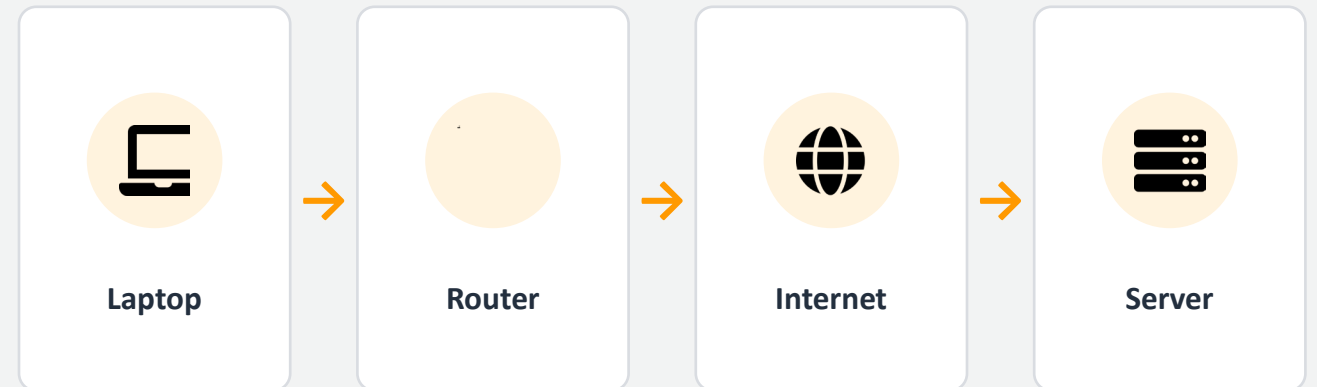
OBJECTIVE: Understand a network as devices connected to exchange information.

A network is two or more devices connected together so they can exchange information.

REAL-WORLD ANALOGY: Like a road system connecting houses to a city, then to other cities.



COMMON MISTAKE: Thinking WiFi and “the Internet” are the same thing — WiFi is just one way to join a network.





Types of Networks

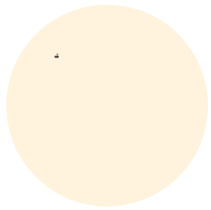
OBJECTIVE: Compare networks by the geographic area they cover.

TYPE		FULL NAME	REAL-WORLD EXAMPLE
	PAN	Personal Area Network	Bluetooth earbuds to phone
	LAN	Local Area Network	Devices in one home or office
	MAN	Metropolitan Area Network	A city's public WiFi network
	WAN	Wide Area Network	A company's offices across states
	Internet	The network of networks	Global — connects every WAN/LAN



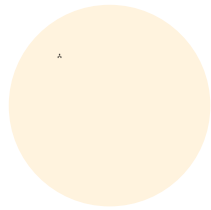
Network Devices

OBJECTIVE: Recognize the hardware that makes networks physically work.



Router

Directs traffic between networks



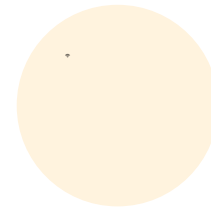
Switch

Connects devices within one LAN



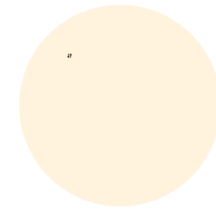
Firewall

Filters traffic for security



Wireless AP

Broadcasts WiFi to devices



Modem

Connects your LAN to your ISP

REAL-WORLD ANALOGY: A modem is your building's mailroom entrance; a router is the front desk directing mail; a switch is the internal hallway connecting offices.



The OSI Model — 7 Layers

OBJECTIVE: See the seven conceptual layers every network transmission passes through.

7	Application	User-facing apps — browsers, email
6	Presentation	Formatting, encryption, compression
5	Session	Opens/manages/closes connections
4	Transport	Reliable delivery — TCP / UDP
3	Network	Addressing & routing — IP
2	Data Link	Node-to-node delivery — MAC, switches
1	Physical	Cables, radio waves, electrical signals



OSI Layers — Analogies & Protocols

LAYER	REAL-WORLD ANALOGY	EXAMPLE PROTOCOLS
7 — Application	<i>Ordering from a restaurant menu</i>	HTTP, HTTPS, FTP, DNS
4 — Transport	<i>The waiter ensuring your full order arrives</i>	TCP, UDP
3 — Network	<i>The delivery address on a package</i>	IP, ICMP
2 — Data Link	<i>The delivery truck's local route</i>	Ethernet, MAC, ARP
1 — Physical	<i>The literal roads the truck drives on</i>	Cables, WiFi radio waves





The TCP/IP Model

OBJECTIVE: See the simplified 4-layer model used in real-world networking.

Application

HTTP, DNS, SSH — combines OSI 5-7

Transport

TCP, UDP — same as OSI Layer 4

Internet

IP — same as OSI Layer 3

Network Access

Ethernet, WiFi — combines OSI 1-2

VS. OSI

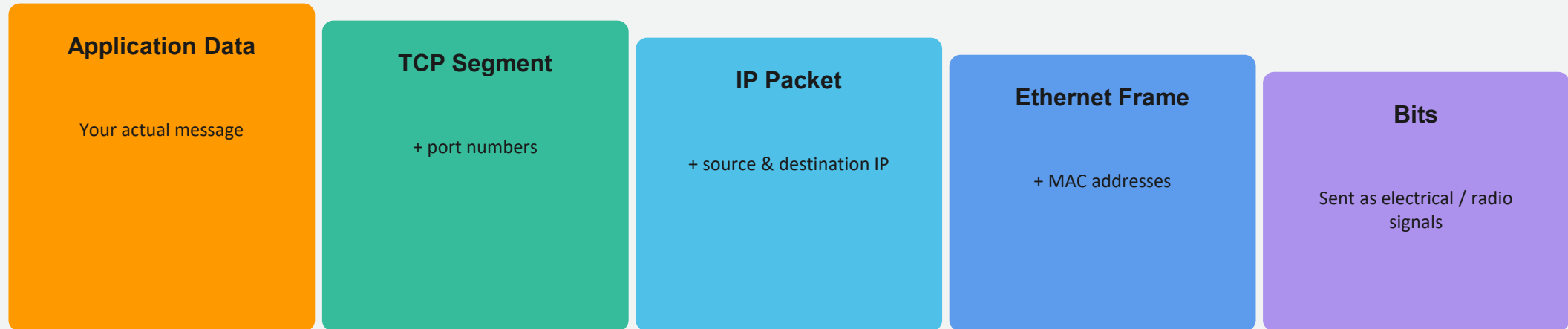
TCP/IP is the practical model the real Internet runs on; OSI is the teaching reference model. Same ideas, fewer layers.

REAL-WORLD ANALOGY: *OSI is the detailed instruction manual; TCP/IP is the quick-start guide engineers use daily.*



Encapsulation — Wrapping Your Data

OBJECTIVE: See how data gets wrapped in layers as it travels down the stack.



REAL-WORLD ANALOGY: Like mailing a letter inside an envelope, inside a padded package, inside a shipping box — each layer adds its own label.



IPv4 Addressing

OBJECTIVE: Understand the structure of a 32-bit IPv4 address.

192.168.1.10

Network portion Host portion

A 32-bit number written as four decimal numbers (0-255), separated by dots — split into a network part and a host part.

EXAMPLES

192.168.1.10

A typical home network device

10.0.0.25

A private AWS VPC instance

172.16.5.1

A corporate private network device



Public vs. Private IP Addresses

OBJECTIVE: Distinguish addresses reserved for private networks (RFC 1918) from public, Internet-routable ones.

PRIVATE (RFC 1918)

Not routable on the public Internet — reused inside millions of private networks.

10.0.0.0/8

16.7 million addresses

172.16.0.0/12

1 million addresses

192.168.0.0/16

65,536 addresses

PUBLIC

Globally unique and routable — how servers are reached directly from anywhere on the Internet.

Google

Public DNS: 8.8.8.8

AWS

EC2 public IPs, Elastic IPs

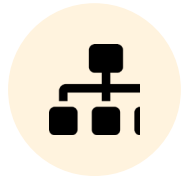
Cloudflare

Public DNS: 1.1.1.1



Subnetting Basics

OBJECTIVE: Break a large network into smaller, organized sub-networks.



Network

Identifies the subnet itself

192.168.1.0



Hosts

Usable addresses for devices

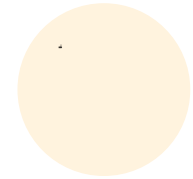
192.168.1.1 - .254



Broadcast

Sends to every device at once

192.168.1.255



Gateway

The router — the “way out”

192.168.1.1

REAL-WORLD ANALOGY: Like a neighborhood: the network is the street name, hosts are house numbers, broadcast is an announcement to the whole street, and the gateway is the road out of the neighborhood.



COMMON MISTAKE: Assigning a device the network address or broadcast address itself — neither can be used as a host address.



CIDR Notation

OBJECTIVE: Read the /24-style shorthand that defines a network's size.

192.168.1.0/24

The number after / is how many bits are the network portion — the rest are for hosts.

CIDR	TOTAL IPs	USABLE HOSTS
/24	256	254
/25	128	126
/26	64	62
/27	32	30
/28	16	14



Hands-On: Splitting a /24

OBJECTIVE: Practice dividing 192.168.1.0/24 into smaller, equal subnets.

2 Subnets

/25

.0/25 and .128/25

126 hosts each

4 Subnets

/26

.0 .64 .128 .192 /26

62 hosts each

8 Subnets

/27

.0 .32 .64224 /27

30 hosts each

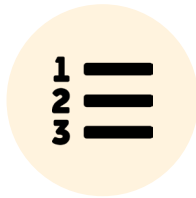


INSTRUCTOR TIP: Trick: each time you split in half, add 1 to the CIDR number and halve the host count.



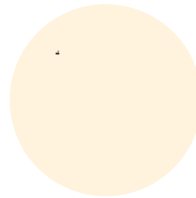
Routing — Getting Packets There

OBJECTIVE: Understand how routers decide where to send each packet.



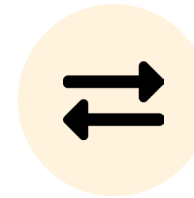
Routing Table

A router's map of known networks & paths



Default Gateway

Where traffic goes if no specific route matches



Packet Forwarding

Moving a packet one hop closer to its destination

REAL-WORLD ANALOGY: Like a GPS: the routing table is the map, the default gateway is “when in doubt, head for the highway,” and forwarding is each turn along the way.



COMMON MISTAKE: Forgetting that a missing or wrong default gateway is one of the most common causes of “no internet access” errors.



Switches vs. Routers

OBJECTIVE: Compare the two devices most often confused by beginners.

CATEGORY	SWITCH	ROUTER
OSI Layer	Layer 2 (Data Link)	Layer 3 (Network)
Purpose	Connects devices within one LAN	Connects different networks together
Uses	MAC addresses	IP addresses
Example	Connecting office desks/PCs	Connecting your LAN to the Internet

REAL-WORLD ANALOGY: A switch is the hallway connecting offices on the same floor; a router is the elevator connecting different floors (networks) entirely.



What Is a Port?

OBJECTIVE: Understand how one IP address can run many different services.

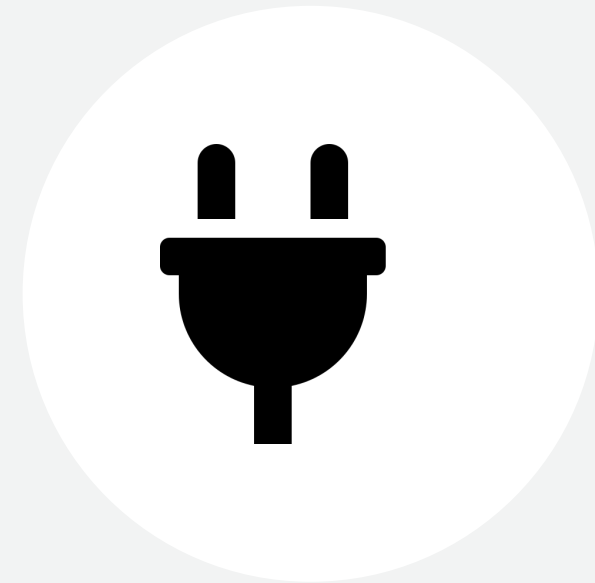
A port is a numbered “door” on a device — it tells traffic exactly which service or application to reach.

IP address = the building

Port = the specific door / suite number



COMMON MISTAKE: Forgetting a firewall can block a port even if the IP address itself is reachable.





Common Ports — Part 1

20 / 21	FTP File transfer	22	SSH Secure remote login	23	Telnet Unencrypted remote login (legacy)
25	SMTP Sending email	53	DNS Domain name lookups	67 / 68	DHCP Automatic IP assignment
80	HTTP Unencrypted web traffic	110	POP3 Retrieving email	123	NTP Time synchronization
143	IMAP Retrieving email (syncd)	389	LDAP Directory services	443	HTTPS Encrypted web traffic



Common Ports — Part 2

445	SMB Windows file sharing	3389	RDP Windows remote desktop	3306	MySQL MySQL database
5432	PostgreSQL PostgreSQL database	6379	Redis Redis cache/database	8080	HTTP-Alt Alternate web port
8443	HTTPS-Alt Alternate secure web port				



INSTRUCTOR TIP: Memory trick: ports under 1024 are “well-known” system ports (like SSH 22, HTTP 80) — databases and custom apps usually live above 1024.

REAL-WORLD ANALOGY: Notice the pattern: HTTPS (443) mirrors HTTP (80); alternates just add “8” in front — 8080 and 8443.



Common Protocols

OBJECTIVE: Recognize the “languages” devices use to communicate.

TCP

Reliable, ordered delivery

UDP

Fast, no delivery guarantee

ICMP

Network diagnostics (ping)

ARP

Finds MAC address for an IP

HTTP

Unencrypted web traffic

HTTPS

Encrypted web traffic

FTP

File transfer

SSH

Secure remote access

SMTP

Sending email

DNS

Domain name lookups



TCP vs. UDP

OBJECTIVE: Compare the two core transport protocols.

TCP

- Connection-based (handshake first)
- Guarantees delivery & order
- Used for: web, email, file transfer
- Slower due to reliability checks

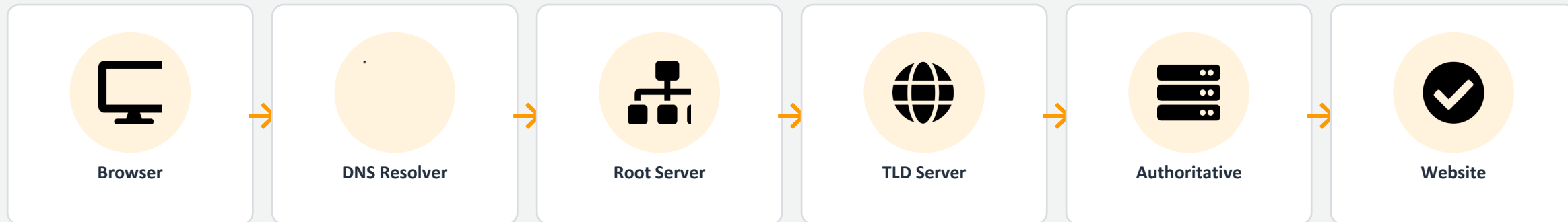
UDP

- Connectionless (no handshake)
- No delivery guarantee — “fire and forget”
- Used for: video calls, gaming, DNS
- Faster due to lower overhead



DNS — The Internet's Phonebook

OBJECTIVE: Trace the full path of a domain name lookup.

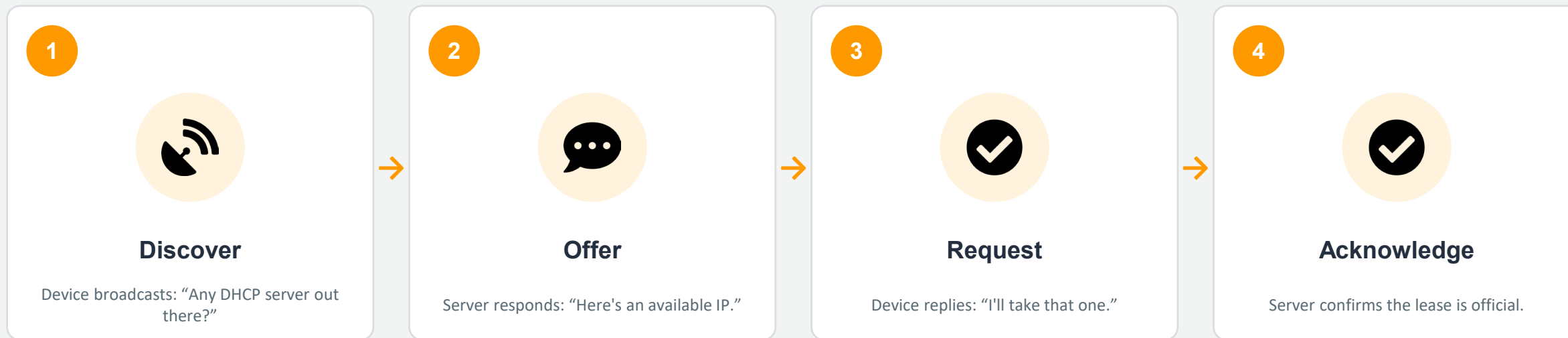


1. Browser asks the DNS Resolver (often your ISP) to look up the domain.
2. Resolver asks a Root Server: “Who manages .com?”
3. Root points to the correct TLD Server for .com domains.
4. TLD points to the domain's Authoritative Server, which holds the real IP.
5. Browser now connects directly to the website using that IP.



DHCP — Automatic IP Assignment

OBJECTIVE: See how a device automatically receives an IP address when it joins a network.



Also known as "DORA" — Discover, Offer, Request, Acknowledge.

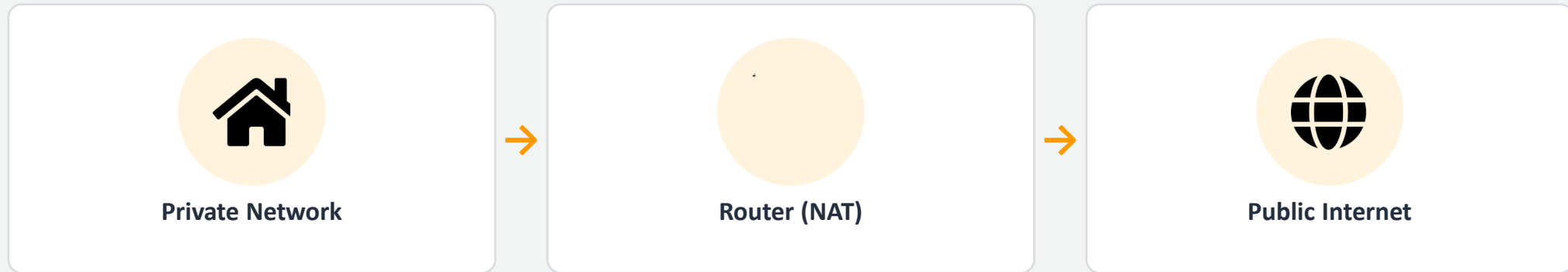


INSTRUCTOR TIP: Recommend the mnemonic "Don't Offer Rotten Apples" (Discover, Offer, Request, Acknowledge) to help it stick.



NAT — Network Address Translation

OBJECTIVE: See how many private devices share one public IP address.



PAT

Port Address Translation — many devices share one public IP, tracked by port

SNAT

Source NAT — rewrites the sender's private IP to a public one

DNAT

Destination NAT — forwards public traffic to a private internal server



Firewalls — Filtering Traffic

OBJECTIVE: Understand how firewalls decide what traffic to allow or block.

Stateful

Tracks the full conversation — automatically allows return traffic for a connection it already approved.

Stateless

Evaluates every single packet independently, with no memory of prior traffic — rules must cover both directions.

REAL-WORLD EXAMPLES

Windows Firewall

Stateful, host-based

AWS Security Groups

Stateful, attached to EC2 instances

AWS NACLs

Stateless, attached to subnets



SECURITY

VPN — Encrypted Tunnels

OBJECTIVE: See how a VPN protects traffic as it crosses an untrusted network.



REAL-WORLD ANALOGY: Like an armored, opaque tunnel through a public highway — outside observers see traffic moving, but can't see or touch what's inside.



INSTRUCTOR TIP: AWS Site-to-Site VPN connects an on-premises network securely to a VPC over the public Internet.



Essential Troubleshooting Commands

OBJECTIVE: Know which command to reach for when something isn't working.

ping

Tests basic connectivity to a host

tracert / traceroute

Shows the path packets take, hop by hop

ipconfig / ifconfig / ip addr

Shows your device's own IP configuration

netstat / ss

Lists active connections & listening ports

arp

Shows local IP-to-MAC address mappings

route

Displays the local routing table

nslookup / dig

Looks up DNS records for a domain

curl

Makes a direct request to a URL

telnet / Test-NetConnection

Tests if a specific port is reachable

Practical Lab — Diagnose Your Network

OBJECTIVE: Run real commands to inspect and test your own network connection.

PowerShell / Terminal

```
$ ipconfig
(view your IP, gateway & DNS server)
$ ping google.com
(test basic connectivity)
$ tracert google.com
(see every hop along the way)
$ nslookup google.com
(confirm DNS resolution)
$ curl google.com
(request the actual webpage)
```

EXPECTED OUTPUTS

- A local IPv4 address & default gateway
- 4 successful replies with a low ms time
- A list of routers between you and Google
- Google's public IP address(es)
- Raw HTML starting with <!doctype html>



INSTRUCTOR TIP: Use Mac/Linux: ping, traceroute, nslookup, curl.
Windows: tracert instead of traceroute.



How This Maps to AWS Networking

OBJECTIVE: See today's concepts become real AWS VPC building blocks.



Internet Gateway = your router's connection to your ISP

VPC = your own private, isolated network (like a whole office building)

Subnet = a floor within that building (with its own CIDR block)

Route Table = the routing rules deciding where each floor's traffic goes

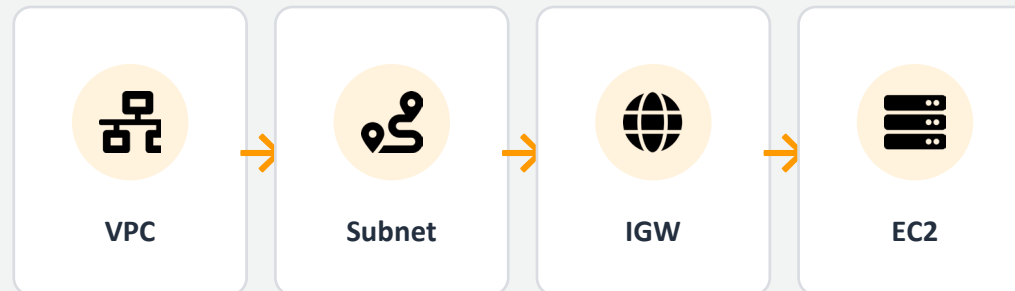


AWS Hands-On Lab — Your First Network

OBJECTIVE: Build a simple, working network in the AWS Console — no coding required.

6 SIMPLE STEPS

- 1 Create a VPC**
CIDR block 10.0.0.0/16
- 2 Create a subnet**
10.0.1.0/24, inside the VPC
- 3 Create & attach an Internet Gateway**
Gives the VPC a path to the Internet
- 4 Update the route table**
Add route 0.0.0.0/0 → Internet Gateway
- 5 Launch a t2.micro EC2 instance**
Place it in your new public subnet
- 6 Connect and test**
SSH into the instance, then ping google.com



You'll Know It Worked When...

Your EC2 instance gets a public IP, you can SSH into it, and it can successfully ping the public Internet.

~30–45 minutes · AWS Free Tier eligible



INSTRUCTOR TIP: Terminate the EC2 instance when you're done — stay inside the Free Tier and avoid surprise charges.



Summary — What We Covered

- ✓ Networks connect devices to exchange data, at different scales (PAN to Internet)
- ✓ OSI & TCP/IP models describe how data is layered and transmitted
- ✓ IPv4 addresses, subnetting & CIDR define how devices are organized and addressed
- ✓ Ports & protocols identify which service traffic is meant for, and how it's delivered
- ✓ DNS, DHCP, NAT, firewalls & VPNs are core services enabling and securing connectivity
- ✓ Troubleshooting tools help diagnose problems layer by layer
- ✓ Every concept maps directly onto AWS VPC networking



COMMON MISTAKES

Avoid These Networking Pitfalls



CIDR Confusion

Thinking a higher CIDR number (e.g. /28) means a bigger network than a lower one (e.g. /24).



Missing Default Gateway

Assuming devices can reach the Internet without a correctly set gateway.



Overly Open Firewalls

Opening all ports “just to make it work” instead of allowing only what's needed.



Confusing NAT with Security

Believing NAT alone makes a private network secure, rather than just address-hidden.



INSTRUCTOR TIP: When troubleshooting, check in order: cable/WiFi → IP config → gateway → DNS → firewall/ports.



Quiz — Questions 1–5

1

What does a router primarily do?

2

Which OSI layer handles physical cables and radio signals?

3

Which model is actually implemented on the real Internet: OSI or TCP/IP?

4

What does the network portion of an IP address identify?

5

Which of these is a private IP range: 8.8.8.8 or 192.168.1.1?



Quiz — Questions 6–10

6 In CIDR notation, does a larger number after the slash mean a bigger or smaller network?

7 What port does HTTPS use by default?

8 Which transport protocol guarantees delivery: TCP or UDP?

9 What type of DNS server holds the actual, final answer for a domain?

10 What does the 'D' in DORA (DHCP) stand for?



Quiz — Questions 11–15

11

What does NAT allow multiple private devices to share?

12

What's the difference between a stateful and a stateless firewall?

13

What does a VPN primarily protect while data crosses an untrusted network?

14

Which command shows the path (hops) packets take to a destination?

15

In AWS, which component plays the same role as a home router's link to your ISP?



Quiz — Answer Key

1. Directs traffic between different networks

2. Layer 1 — Physical

3. TCP/IP

4. Which network a device belongs to

5. 192.168.1.1 (private)

6. Smaller network

7. Port 443

8. TCP

9. Authoritative server

10. Discover

11. One public IP address

12. Stateful remembers connections; stateless checks every packet

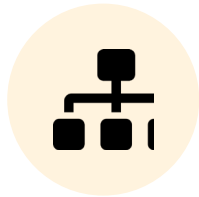
13. The content/data itself

14. traceroute / tracert

15. Internet Gateway

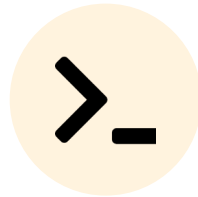


Homework



Subnet Practice

Split 10.0.0.0/16 into 4 equal subnets and write out each CIDR block.



Command Practice

Run all 5 lab commands against 3 different websites and record the results.



Read: AWS VPC Docs

Skim the AWS VPC “Getting Started” guide to preview tomorrow's hands-on lab.

Reference & Cheat Sheets

Quick-reference material to keep handy as you continue building AWS skills.





Networking Cheat Sheet

IP Address

Unique identifier for a device on a network

Subnet

A smaller, segmented piece of a larger network

CIDR

Shorthand for network size, e.g. /24

Gateway

The router that connects a subnet to other networks

DNS

Translates domain names into IP addresses

DHCP

Automatically assigns IP addresses to devices

NAT

Lets many private devices share one public IP

Firewall

Filters traffic based on rules

VPN

Encrypts traffic across an untrusted network

Port

Identifies which service traffic is meant for

Router

Connects different networks together

Switch

Connects devices within the same network



Port Cheat Sheet

20/21 FTP	22 SSH	23 Telnet	25 SMTP	53 DNS	67/68 DHCP
80 HTTP	110 POP3	123 NTP	143 IMAP	389 LDAP	443 HTTPS
445 SMB	3306 MySQL	3389 RDP	5432 PostgreSQL	6379 Redis	8080/8443 HTTP/S Alt



INSTRUCTOR TIP: 22=SSH, 80/443=Web, 53=DNS — memorize these three first; the rest follow with repetition.



CIDR Cheat Sheet

CIDR	TOTAL IPs	USABLE HOSTS
/16	65,536	65,534
/20	4,096	4,094
/24	256	254
/25	128	126
/26	64	62
/27	32	30
/28	16	14
/29	8	6
/30	4	2



OSI Cheat Sheet

7	Application	HTTP, DNS, FTP
6	Presentation	SSL/TLS, encoding
5	Session	Session establishment
4	Transport	TCP, UDP
3	Network	IP, ICMP
2	Data Link	Ethernet, MAC, ARP
1	Physical	Cables, radio waves

Mnemonic: Please Do Not Throw Sausage Pizza Away (bottom → top)



TCP vs. UDP — Quick Reference

CATEGORY	TCP	UDP
Connection	Connection-based (handshake)	Connectionless
Reliability	Guaranteed delivery & order	No guarantee
Speed	Slower (more overhead)	Faster (less overhead)
Common Uses	Web, email, file transfer	Video calls, gaming, DNS



Useful Commands Reference

ping <host>

Test connectivity

tracert/tracert <host>

Trace the path taken

ipconfig / ifconfig / ip addr

Show local IP config

netstat -an / ss -tulpn

List active connections

arp -a

Show local ARP table

route print / ip route

Show routing table

nslookup <domain>

Basic DNS lookup

dig <domain>

Detailed DNS lookup

curl <url>

Request a URL directly

telnet <host> <port>

Test if a port is open

Test-NetConnection <host>

Windows port/connectivity test



Great Work Today!

You now understand how networks, IP addressing, and the Internet itself actually work — the foundation for everything in AWS VPC.